

3. CLO 3. (5 คะแนน) Prove: Let a , b , and c be integers. If $(a - b) \mid c$, then $a \mid c$.

Remember that you must use the definition of x divides y ($x \mid y$) (Definition 1.1) in your proof.

let a, b and c are integers, that $(a - b) \mid c$ then also

by Definition 1.1 assume $(a - b) \mid c$ is true then

we know that $(a - b) \mid c$ is $(a - b) \cdot k = c$, $a \cdot b \cdot k = c$, $ak_1 = c$

that $x \mid y = y = x \cdot k$ so $ak_1 = c$ is also true there is true

7. (10 คะแนน) Prove the following statement by contraposition.

Let x be an integer. If $x^2 + x + 1$ is even, then x is odd.

Make sure that your proof makes appropriate use of

Definitions 1.2 (about even numbers) and 1.3 (about odd numbers) discussed in class.

determine x is an integer to assume if $x^2 + x + 1$ is even, then x is odd

from $x^2 + x + 1$ write as $x(x + 1) + 1$ we know that if x is odd so $x + 1$ is even

so by Definition 1.2

then $x(x + 1) = 2k$

by Definition 1.3 we found that $x(x + 1) = 2k$

then $x(x + 1) + 1 = 2k + 1$

so $x^2 + x + 1$ can't be even and we can't know that x is odd there is false

11. CLO 3. (20 คะแนน) Write a proof by contradiction of the following.

Let x and y be integers. If x and y satisfy the equation

$$3x + 5y = 153$$

then at least one of x and y is odd.

Remark: โจทย์ข้อนี้เคยถูกเลือกเป็นข้อสอบกลางภาค

let x and y are integers that if $3x + 5y = 153$ then at least one of x and y is odd

assume x and y are even both then

by Definition 1.2

$x = 2k_1$ and $y = 2k_2$ while k_1 is integer.

then $6k_1 + 10k_2 = 153$

$$2(3k_1 + 5k_2) = 153$$

but $\frac{153}{2}$ is not in integer by integer attribute if x and y are integers operation multiply and plus must give a result

that be in integer so

"if $3x + 5y = 153$ then at least one of x and y is odd" is true.

13. CLO3. (15 คะแนน) Let " $x \nmid y$ " denote " x does not divide y ." Prove by contraposition.

Let a and b be integers. If $5 \nmid ab$, then $5 \nmid a$ and $5 \nmid b$.

$x \nmid y$ And a and b are integers that if $5 \nmid ab$ then $5 \nmid a$ and $5 \nmid b$

Assume $5 \nmid a$ or $5 \nmid b$ then $5 \nmid ab$

that we know by Definition 1.1 $a = 5k_1$ $b = 5k_2$

for $5 \nmid a$ then $5 \nmid ab$

$$ab = 5k_1b \text{ and}$$

for $5 \nmid b$ then $5 \nmid ab$

$$ab = 5k_2a \text{ and}$$

for $5 \nmid a, 5 \nmid b$ then $5 \nmid ab$

$$ab = 5k_1, 5k_2$$

So every statement is true then

"if $5 \nmid ab$, then $5 \nmid a$ and $5 \nmid b$ " is true.

16. CLO 3. (20 คะแนน) Consider the following definitions.

Definition. An integer n is alphic if $n = 4k+1$ for some integer k .

Definition. An integer n is gammic if $n = 4k+3$ for some integer k .

(a) (5 คะแนน) Show that 19 is gammic.

(b) (5 คะแนน) Suppose that x is alphic and y is gammic. Prove that $x + y$ is even.

(c) (10 คะแนน) Prove by contraposition: If x is not odd, then x is not alphic.

17. CLO 3. (5 คะแนน) Prove that the rational numbers are closed under multiplication. That is, prove that, if a and b are rational numbers, then $a \cdot b$ is a rational number.

Definition a rational number is a number that can be written as a/b , where a and b are integers with $b \neq 0$

18. CLO 3. (25 คะแนน) Use Proof by Contraposition to show that if x and y are integers and $x^2(y^2 - 2y)$ are odd then both x and y are odd.

16. (a) let x is 19 and x is gammic by Definition

then $19 = 4(4)+3$ so 19 is gammic

(b) let x is alphic and let y is gammic

by Definition and Definition 1.3

we can assume $4k+1$ as $2k_1+1$ while $k_1 = 2k$

and assume $4k+3$ as $2k_1+2k_2+1$ while $k_2 = 1$

$2k_1+1$ is odd then from $2k_1+1+2k_2$ is odd plus even then the result would be odd

so that x and y are odd either so odd plus odd is $2k_1+1+2k_2$ equal to $4k+2$ is even

so $x \cdot y$ is even is true.

continue

16 (c). let x isn't odd then x isn't alphic

Assume x is alphic then x is odd

by 16(b) we know that any alphic is odd so x is odd then

" x isn't odd then x isn't alphic" is true.

19. Let a and b are rational numbers then $a \cdot b$ is rational number

by Definition let c_1 and d_1 are integers then and $d_1 \neq 0$

$$a = c_1/d_1, \quad b = c_2/d_2$$

$$\text{then } a \cdot b = \frac{c_1}{d_1} \cdot \frac{c_2}{d_2} = \frac{c_1 c_2}{d_1 d_2} = \frac{x}{y}$$

while $y = d_1 d_2$ then $y \neq 0$ so $\frac{x}{y} = w$ is rational number.

conclude a, b are rational numbers then $a \cdot b$ is rational number

18. CLO 3. (25 คะแนน) Use Proof by Contraposition to show that if x and y are integers and $x^2(y^2 - 2y)$ are odd then both x and y are odd.

let x and y are integers if $x^2(y^2 - 2y)$ is odd then x and y are odd.

Assume x or y is/are even then $x^2(y^2 - 2y)$ is even.

by Definition 1.2 for x is even then $x = 2k$

$$\text{so } (2k)^2(y^2 - 2y) = 4k^2 y(y - 2) \text{ is } 2k \text{ which } k = 2k y(y - 2)$$

So x is even then $x^2(y^2 - 2y)$ is even is true.

Also $x^2(y^2 - 2y)$ is odd then x and y are odd is true.